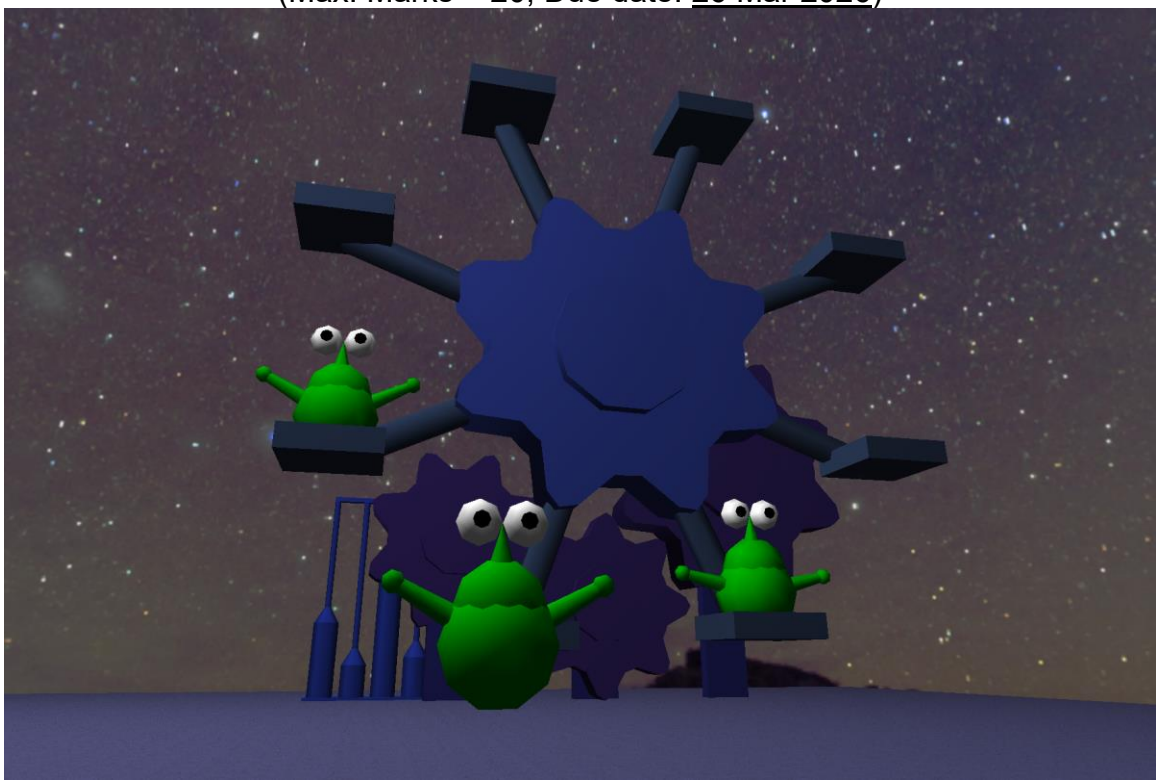
COSC363: Computer Graphics
Assignment 1

Alien Theme Park

(Max. Marks = 20; Due date: 26 Mar 2026)



1. The Scene

Aliens wander around a set of one or more moving theme park rides.

2. Minimum Requirements (Max. 7 Marks)

In this assignment, you will develop an OpenGL program to display your version of the scene as described in the text prompt above. The minimum requirements are outlined below.

2.1 A model of a theme park ride which displays some kind of animation (e.g. a rotating Ferris wheel or pendulum). This need not be made of highly complex models. It could be constructed using a set of GLU/GLUT objects.

2.2 Animated aliens: Alien models may also be constructed using GLU/GLUT objects. They need not necessarily have humanoid characteristics. Each alien should perform some action such as continuously moving along a path, waving their arms while on one of the theme park rides etc.

2.3 Lighting: The scene should have proper lighting so that the models and animations are clearly visible.

2.4 Camera: The user should be able to move the camera through the scene using a set of directional keys: E.g. Up arrow: move camera forward in the current direction, Down arrow: move camera backward in the current direction, Left arrow: Turn left by 5 degs, Right arrow: Turn right by 5 degs.

2.5 Textures: At least three different textures must be used in the scene. It is not necessary to texture-map all surfaces. Please do not use very large images (>10MB) as textures. A skybox texture set, if used, will be counted as a single texture.

2.6 Wireframe mode: The user should be able to toggle a wireframe mode on/off by pressing a key (e.g. 'q'). When wireframe mode is on, polygon filling and lighting must be disabled.

3. Extra Features and Marks (Max. 8 Marks)

The assignment will be marked out of 20, with 7 marks allocated for minimum requirements, 3 marks for the overall rendering quality, and 2 marks for a report (see below). With just the basic features given in 2.1–2.6 above, your assignment would earn at most 12 marks out of 20. You will need to add a few more features to the scene to get additional marks. Some possible features, and the marks they would each gain if implemented correctly, are listed below.

3.1 Planar shadows cast by at least two objects (1 mark)

3.2 Complicated pathing: the path of a theme park ride's seats must be at least more complicated than a smooth linear or elliptical path with fixed speed. A physics-based pendulum or curving (but not elliptical) train track would count (up to 2 marks). Any equations used must be supplied in the report.

3.3 Aliens get on and off one of the theme park rides. They must not teleport around to achieve this. (1 mark)

3.4 A sky box or sky dome/sphere (1 mark)

3.5 A dynamic spotlight: it must either be attached to a moving/rotating object, or be changing colour/intensity over time. The movement/colour/intensity change of the spotlight should be clearly visible (1 mark).

3.6 Smooth user movement: all four movement controls (turn left and right, move forward and backward) must be changed to work smoothly. For example, if the backward key is pressed down, the camera should immediately start moving backwards at a fixed and steady rate until the key is released (0.5 marks).

3.7 A custom-built sweep surface (a surface of revolution or extrusion) constructed using quad strips (up to 3 marks). This will require evidence in the report detailing

how the surface was generated. It cannot be a shape that can be created with built-in GLU/GLUT functions (e.g. cylinder, sphere, cone). Must have correct normals for full marks.

3.8 A dynamic mesh: it must be animated with deforming geometry which is clearly visible (especially in wireframe mode), and which is not achievable with scaling/rotation/translation of the whole object (e.g. a bouncing trampoline). More marks will be given based on how well the mesh deformation fits the rest of the scene (up to 3 marks)

3.9 Particle systems: more marks will be given based on the complexity of the system. Simply implementing a particle system similar to that in Lab 5 will only gain you 1 mark out of 3. More marks will be awarded for features (which must be specified in the report) such as: turning on/off periodically, having bouncing particles, fading out/culling particles as they approach the floor plane, etc. (up to 3 marks)

The marks associated with each feature are indicative of the time and/or effort required to implement that feature.

4. Rendering Quality (Max. 3 Marks)

In the context of this assignment, the term "rendering quality" refers to the visual aesthetic quality of the models and animations displayed by the program. Slow and jittery animations, incorrectly rendered models and shadows, improper movement of the camera, and improper illumination may cause marks to be deducted for poor rendering quality.

You will gain marks only for models/animations that are correctly and clearly displayed on the screen. A particular feature may be implemented correctly in code, but if the corresponding output is not clearly seen on the screen, you will not receive marks for that feature.

5. Report (Max. marks: 2, Max. number of pages: 4)

The report should include the following:

- Your name and student number
- A brief description of the scene (1 paragraph).
- At least two screenshots showing important aspects of the scene or animation
- An itemised list of *all* extra features implemented.
- Images of diagrams, sketches used for designing models (e.g. the base curve of a sweep surface).
- Equations if any, used for designing surfaces and animations.
- A full list of control functions (keyboard, mouse, special keys) defined for interacting with the scene.

- Build commands or instructions for compiling and running the program on CSSE lab machines. If the program was developed using Visual Studio on a Windows platform, please mention that in your report.
- References to external sources of textures, models, algorithms, equations etc., used in your work.
- A statement of acknowledgement if you used generative AI (see section 8).
- **Important!** The report must contain the following declaration. **Assignment submissions without the declaration will not be accepted**.

Declaration
I declare that this assignment submission represents my own work (except for allowed material provided in the course), and that ideas or extracts from other sources are properly acknowledged in the report. I have not allowed anyone to copy my work with the intention of passing it off as their own work. Name: Student ID: Date:

You may include more than 4 pages in the report, only if absolutely necessary. Please submit your report as a single **PDF** file.

6. Models and Animations

The object models in the scene may be constructed using any of the following methods:

- By transforming and combining a set of (at least four) GLUT/GLU objects such as spheres, cubes, cylinders etc. to form a composite model.
- By using a set of polygonal surfaces comprising of quads and triangles.
- By using surface generation methods such as sweep surfaces, surfaces of revolution etc.

You are not required to design highly complex mesh models for this assignment. Using objects modelled using design packages such as 3DS Max, Blender, Lightwave, etc., or downloading pre-built mesh models from the Internet will not give you any extra points.

If you use any external model files (e.g. loading OFF or OBJ files), you must make sure the license permits usage, and reference the source in your report. Please keep in mind that using external models alone will not give you any marks for the modelling related criteria (e.g. theme park ride model, animated alien models). You will need to meet these criteria by creating your own geometry (which could contain external models as a subpart) within your assignment code.

You may use parts of lab code and resources (models, images, image loading functions) provided in the course. Models and animations developed in the lab will give you marks only if significant changes or enhancements have been made to them.

7. Program Development:

- Please do not use source codes of programs from online repositories, tutorials, and books. The one exception is that you may use an external, header-only OBJ file loader if the license permits usage. If used, this must be appropriately referenced in the report.
- Develop your program in C/C++ language using only OpenGL 2 API. Please do not use OpenGL 4 code (vertex/fragment shaders etc.), or extensions (e.g. ARB, EXT etc.) that are not part of the standard OpenGL-2 API.
- **Please make sure that your program can be compiled and run on CSSE lab machines.**
- Your submission will not be marked for code readability or optimization. However, you are encouraged to prepare well documented and readable code.

8. Generative AI Usage



Generative AI Tools Are Permitted for Certain Parts of This Assessment

In this assessment, you are permitted to use generative artificial intelligence (AI) solely for the purposes of explaining the usage of computer graphics functions, answering questions about debugging, generating short code segments for reuse (1-3 lines), and grammar checking. No other use of generative AI is permitted. To assist with maintaining academic integrity, you must appropriately acknowledge any use of generative AI in your work. Please include a statement of acknowledgement with your work, clearly indicating which AI tools were used and how they contributed to your assessment.

9. Assignment Submission

Submit electronically (using *Learn*), the source code and all supplementary files (models, images etc.) needed to run the program. Please also submit your report in PDF format. The files should all be packaged together and submitted as a single .zip file.

This is not a group project. Your assignment submission must represent your own individual work. In particular, students are not permitted to share program source code in any way.

10. Miscellaneous

1. Check regularly on the *Learn* system forums for spec updates and clarifications.
2. Standard departmental regulations regarding dishonest practices apply.